

Lab: Explore Prompt Writing



Estimated Time: **15** minutes

Objectives

- Compare and contrast prompt writing approaches
- Choose a prompt approach based on the goal of your interaction with the AI
- Write a prompt using key elements for writing effective prompts and the prompt writing process

Prerequisites

You should be familiar with:

- Elements of good prompts
- Prompt writing process
- Approaches to writing prompts

Introduction

In this lab, you will explore writing different prompts based on a scenario we provide. We will provide you with prompt choice elements. You will mix and match these prompt elements to form similar but varying prompts so you can compare and contrast outputs based on the elements you chose.

Scenario

A director of a small non-profit organization dedicated to working with its community to improving the longevity and quality of life of its members. They want to launch a new program educating seniors on how to sensibly use AI. She has heard the term "AI hallucinations" and wants to understand the term better to see if it's the program needs to address the topic.

Review

You can choose to skip this **Review** section and continue to the **Practice** section if you feel comfortable with the prerequisites.

Elements of good prompts

- Define a clear goal for the interaction
- Avoid ambiguity, write clearly and concisely
- Provide context
 - Specify output expectations:
 - Specify desired output structure

- Identify constraints for the output
- Provide a point of view / persona
- Ask the AI to state its assumptions

Prompt writing approaches

Interaction Purpose	Description	Approaches
Information retrieval	Factual knowledge	Zero-shot, One-shot, Few-shot, Interview
Analysis and reflection	Insight generation, exploration, logical analysis	Chain-of-thought, Self-reflection, Tree-of-thought
Creative idea generation	Brainstorming	Role-play, Scenario simulation
Instructional guidance	Sequential, structured output	Process decomposition, Templating, Guided narrative
Collaborative exploration	Open-ended, unpacking unfamiliar or complex topics	Conversational prompting, Iterative refinement

Prompt writing process

1. Define your goal, including the purpose of your interaction. Your purpose will help you decide on which approach(es) will help you meet your goal.
2. Write and test the prompt
3. Analyze the response
4. Iterate

Practice

In this guided practice exercise, you will explore writing different prompts to learn more about the term "AI hallucination" from the perspective of the scenario provided at the beginning of the lab.

Recall the scenario

A director of a small non-profit organization dedicated to working with its community to improving the longevity and quality of life of its members. They want to launch a new program educating seniors on how to sensibly use AI. She has heard the term "AI hallucinations" and wants to understand the term better to see if the program needs to address the topic.

Instructions

You will construct prompts from prompt elements we provide you. You will mix and match to compare outputs from similar, but varying, prompts.

Step 1. Goal choice

Topic: AI hallucinations

Goal: Determine the depth to which the topic of AI hallucinations needs addressed in the provided scenario.

Step 2. Write and test the prompt

In this step, you will see:

- Sample prompt questions
 - Context descriptions
 - Output expectations
1. Choose one element from each section to construct your prompt, then enter it into the *AI Classroom* tool.
 2. Copy and paste the prompts and the responses into a text document so you can compare them.

Sample prompts (Choose one)

- Define the term "AI hallucination." (Information retrieval)
- Why do AI models hallucinate? (Analysis)
- Provide four examples of widely reported hallucinations. Describe the factors that made these hallucinations particularly noteworthy. (Analysis, information retrieval)
- What contexts would motivate seniors to learn the term "AI hallucinations"? (Idea generation)
- What causes hallucinations? Explain the steps you used to reach this conclusion. (Instructional guidance)
- Why would a senior community member need to understand the term "AI hallucination"? (Collaborative exploration)

Possible contexts (Choose one)

- I'm developing a curriculum to educate seniors in my community on how to use AI so they can use AI wisely.
- I'm working with my community to improve the longevity and quality of life of its members.
- As a senior consumer wanting to use AI, I want to better understand the term "AI hallucinations".

Output expectations (Choose at least one)

- Do not include an output expectation
- Explain why you chose this approach
- Identify potential assumptions
- Provide your references

Example prompt

- Sample prompt 3
- Context 1
- Output expectation 4

Provide four examples of widely reported hallucinations. Describe the factors that made these hallucinations particularly noteworthy.

I'm developing a curriculum to educate seniors in my community on how to use AI so they can use AI wisely so I would like to use these factors to help them understand why this is an important concept when using AI.

Provide your references.

Step 3. Analyze

Evaluate how well the output aligns with your goal.

Consider:

- Is there any information missing
- Is the response free from bias?
- Do you need a more conceptual response or a specific response?
- Do you want a different point of view?

Step 4. Iterate

After considering the output, either revise and refine the entire prompt and try again or further investigate an aspect of the output.

For further exploration

Choose your own topic to investigate with AI

- Decide on your goal
- Write and test the prompt
- Word your inquiry using concise phrasing while also providing sufficient information to achieve your objective
- Provide appropriate context
- Specify your output expectations, including structure, constraints, persona, and bias considerations for the output

Evaluate

Consider how well the output aligns with your goal:

Is there any information missing

Is the response free from bias?

Do you need a more conceptual response or a specific response?

Do you want a different point of view?

Iterate

After considering the output, either revise and refine the entire prompt and try again or further investigate an aspect of the output.

Congratulations!

You have completed this lab. You now have experience with:

- Comparing and contrasting prompt writing approaches
- Choosing a prompt approach based on the goal of your interaction with the AI
- Writing a prompt using key elements for writing effective prompts and the prompt writing process

Next Steps

You can apply the prompt techniques from this lab to real world needs. Return to this lab if needed to revisit these techniques.

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